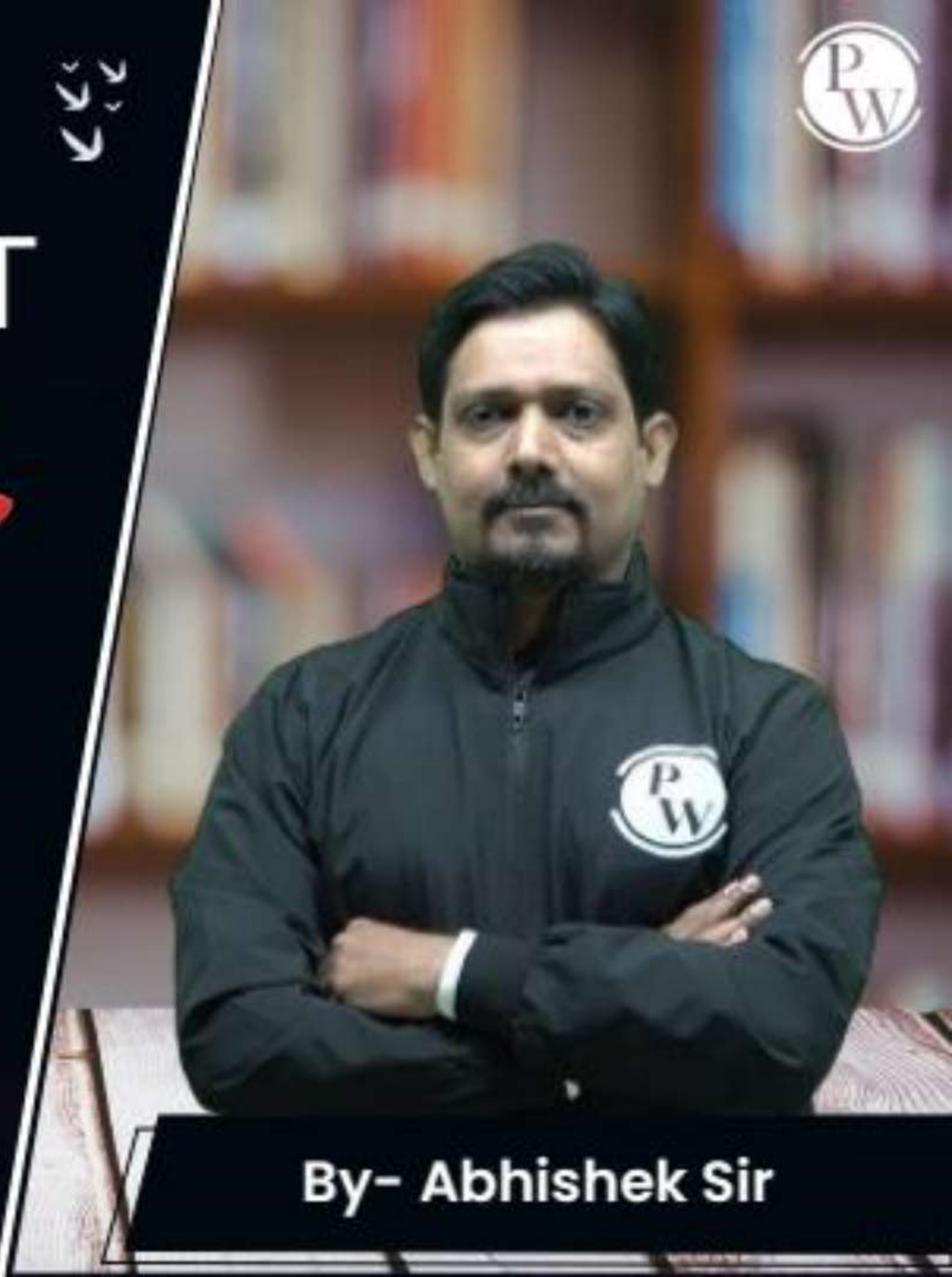
Computer Science & IT



C Programming

Language Processor

Lecture No. 02



By- Abhishek Sir

Recap of Previous Lecture



Topic — Introduction

Topic — Language Translator

Topic — Syllabus & Weightage

Topic

Topic

3 Question

Slide

Topics to be Covered



- Topic — Data Types
- Topic — Integer
- Topic — Character
- Topic
- Topic

Slide

primitive

Derived data type

user defined data



Topic : Data Types

- Primitive
- Derived
- User Defined



Topic : Primitive

1. Integer ✓
2. char ✓
3. float ✓ → 4Byte Single precision
   double 8 byte Double precision.

Slide



Topic : Derived

Data type

* Array ✓
* String (Array)
* pointer

Slide



Topic : User Defined

* Structure (collection of dissimilar element)
* union
* enum ✓
* typedef (alias another name)

Student : Name
Roll No
phone No.

Slide



Topic : Declarative Statement

Integer

int a; ← declarative statement

Keyword

Reserve

Name of variable

Variable - (identifier)

Slide

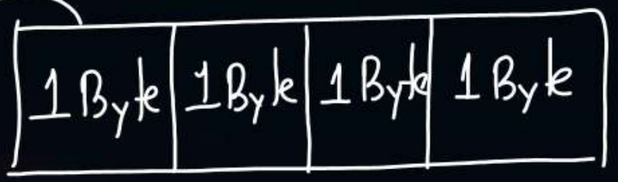


Topic : variable

int = 4 Byte Size (windows)

Compiler will allocate 4 Bytes of memory. (Main Memory)

address :-



1000

assume

a

4B

4 Bytes = 32 bits

1 byte = 8 bits

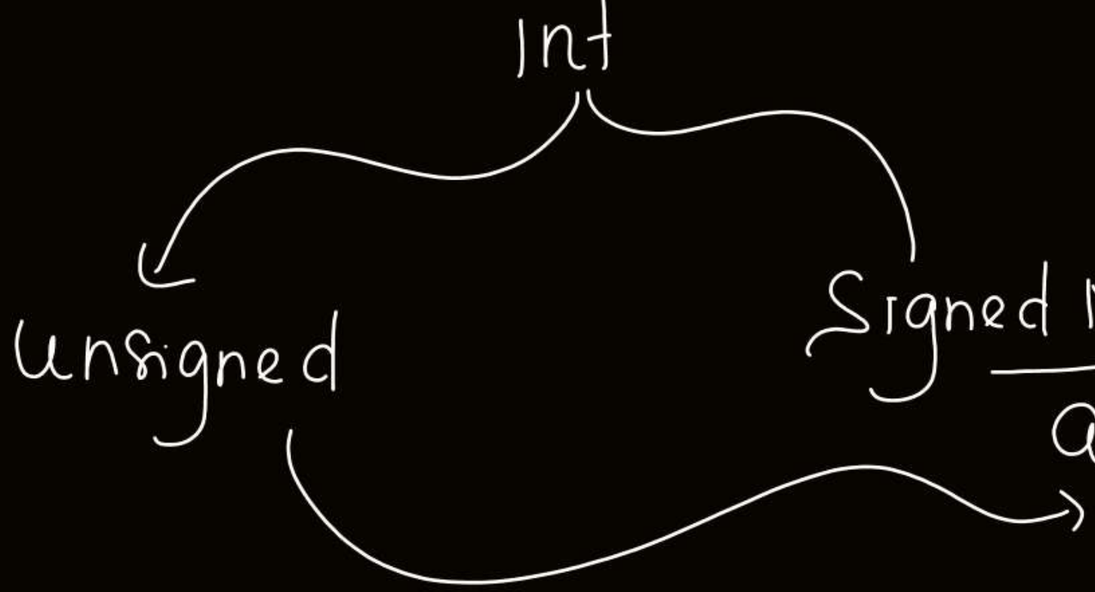
Slide

int a;
a = 10; ← Expression.

Data

int
- short int 2Byte
- int 4B
- long int 8B
- long long int.

} Amount of memory.



int - 2 Byte (Assume)

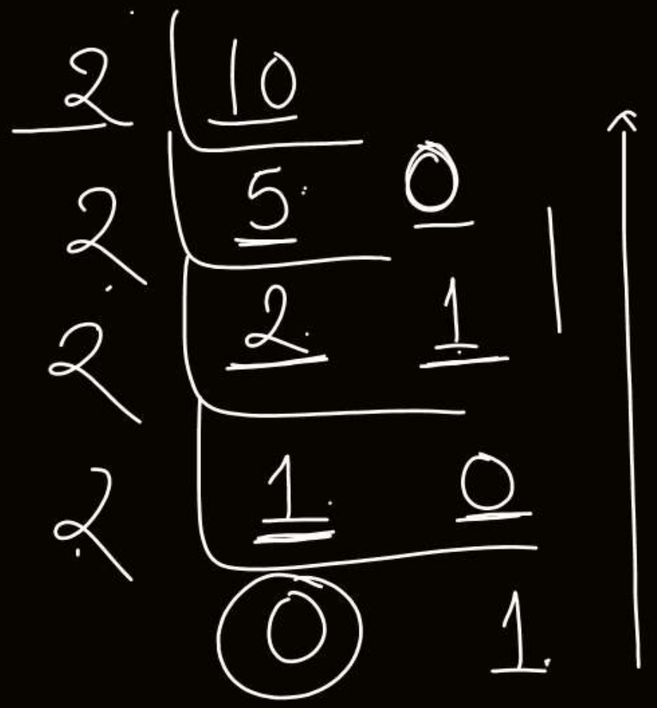
Binay Representation

Unsigned int a = 10;

Decimal No ←

Human Comfortable - Decimal

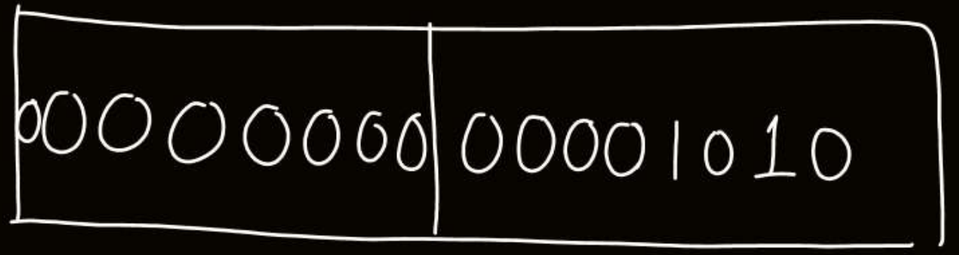
0, 1, 2, 3 .. 9.



1010 ← Binary Representation

↑ unsigned No

Size of integer = 2B



unsigned No

Range of unsigned No

1 0 1 0 ← Binary

3 2 1 0

$1\times2^3 + 0\times2^0 + 1\times2^1 + 0\times2^0$

$8 + 0 + 2 + 0 = 10$

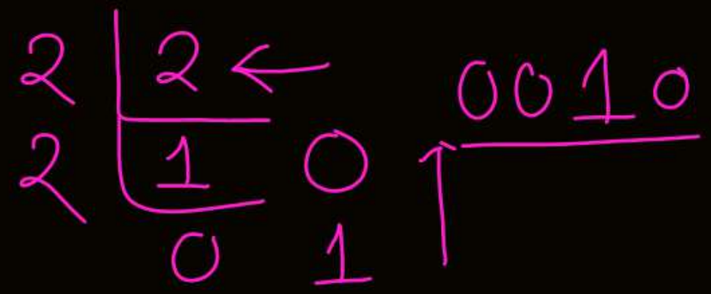
Assume Size of int = 2 Byte

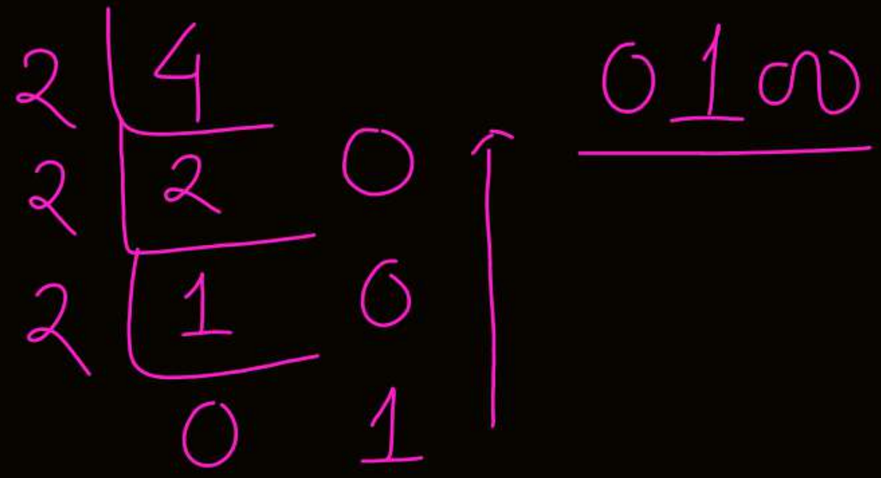
0000	0 ←
0001	(1)
0010	(2)
0011	(3)
0100	(4)
0101	(5)
0110	(6)
0111	(7)

1000	8
1001	9
1010	10
1011	11
1100	12
1101	13
1110	14
1111	15 ←

4bit 0 —— 15 (4bits)

0 — $2^4 - 1$





decimal

000 ←

0 — 999

00

ggg

gg

Unsigned

4bit

0 0 0 0

$0 \text{ — } 2^4 - 1$

8 4 2 1

1 1 1 1

3 2 1 0

$1\times2^3 + 1\times2^2 + 1\times2^1 + 1\times2^0$

$8+4+2+1 = 15$

8bit $0 \text{ — } 2^8 - 1$ → 255

2Byte(short) $0 \text{ — } 2^{16} - 1$ → 0-65535

int (4Byte) $0 \text{ — } 2^{32} - 1$

long int (8byte) $0 \text{ — } 2^{64} - 1$

$2^0 = 1$

$2^1 = 2$

$2^2 = 4$ ←

$2^3 = 8$

$2^4 = 16$

$2^5 = 32$

$2^6 = 64$

$2^7 = 128$

$2^8 = 256$

$2^9 = 512$

$2^{10} = 1024$

$2^{11} = 2048$

$2^{12} = 4096$

$2^{13} = 8192$

$2^{14} = 16384$

$2^{15} = 32768$

$2^{16} = 65536$ ✓

Memorize

n bit unsigned Range = $0 - 2^n - 1$ ←

int a = -10; integer can be Negative

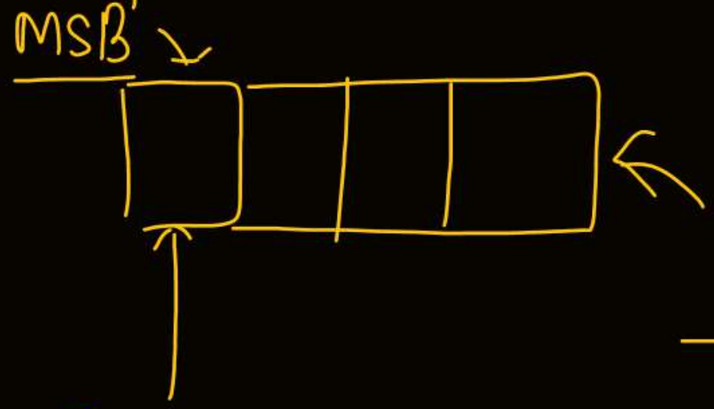
programming ~~Sig~~ Language

Sign No. represented by 2's complement representation

2's complement

10 ⟶ 1010

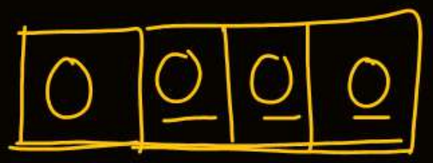
Sign — bit pattern's MSB — ~~Sign~~ represent the Sign



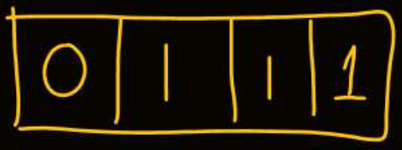
Most significant bit

MSB → 0 +ve No

MSB · 1 - -ve No



+Ve No



↑

0 ——— 7

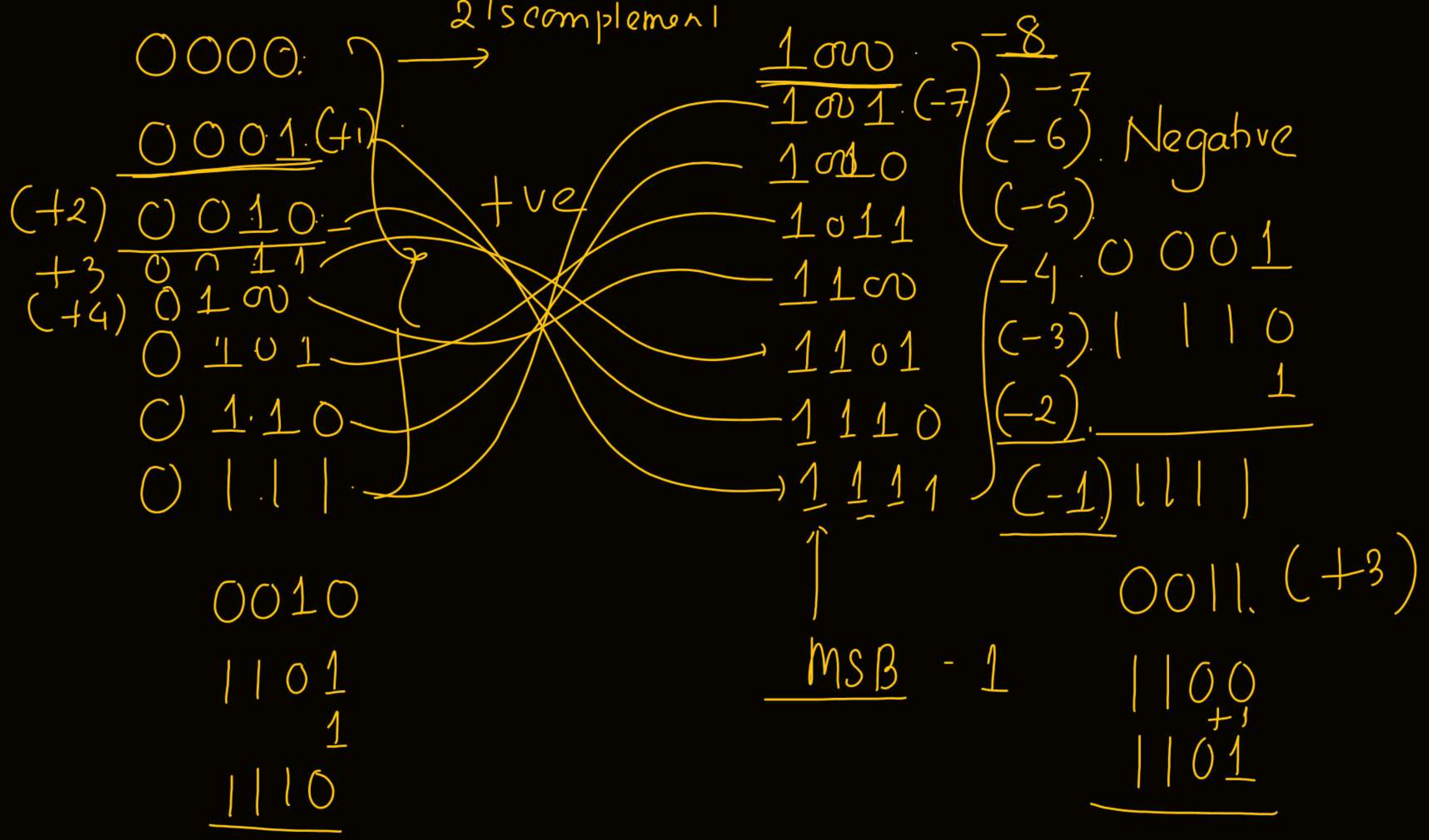
(+1) 0001 ⟶ 2's complement

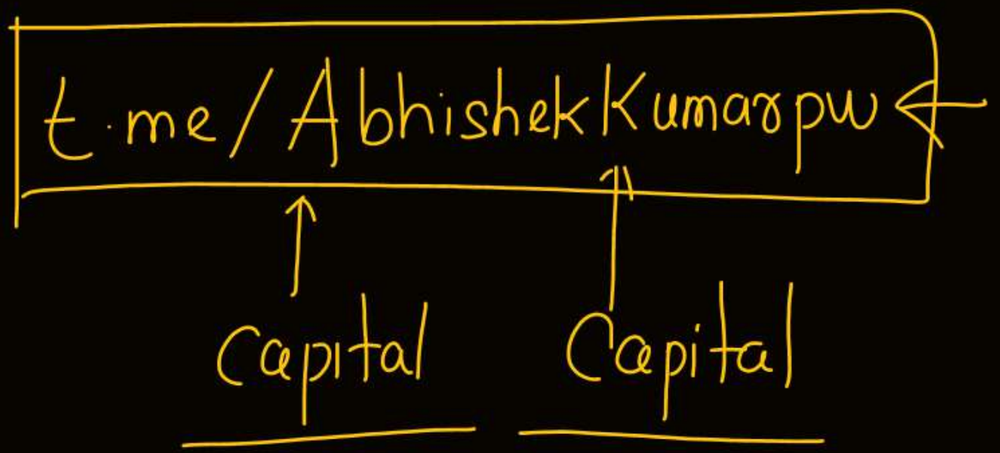
(1'S complement ~~and~~ add 1)

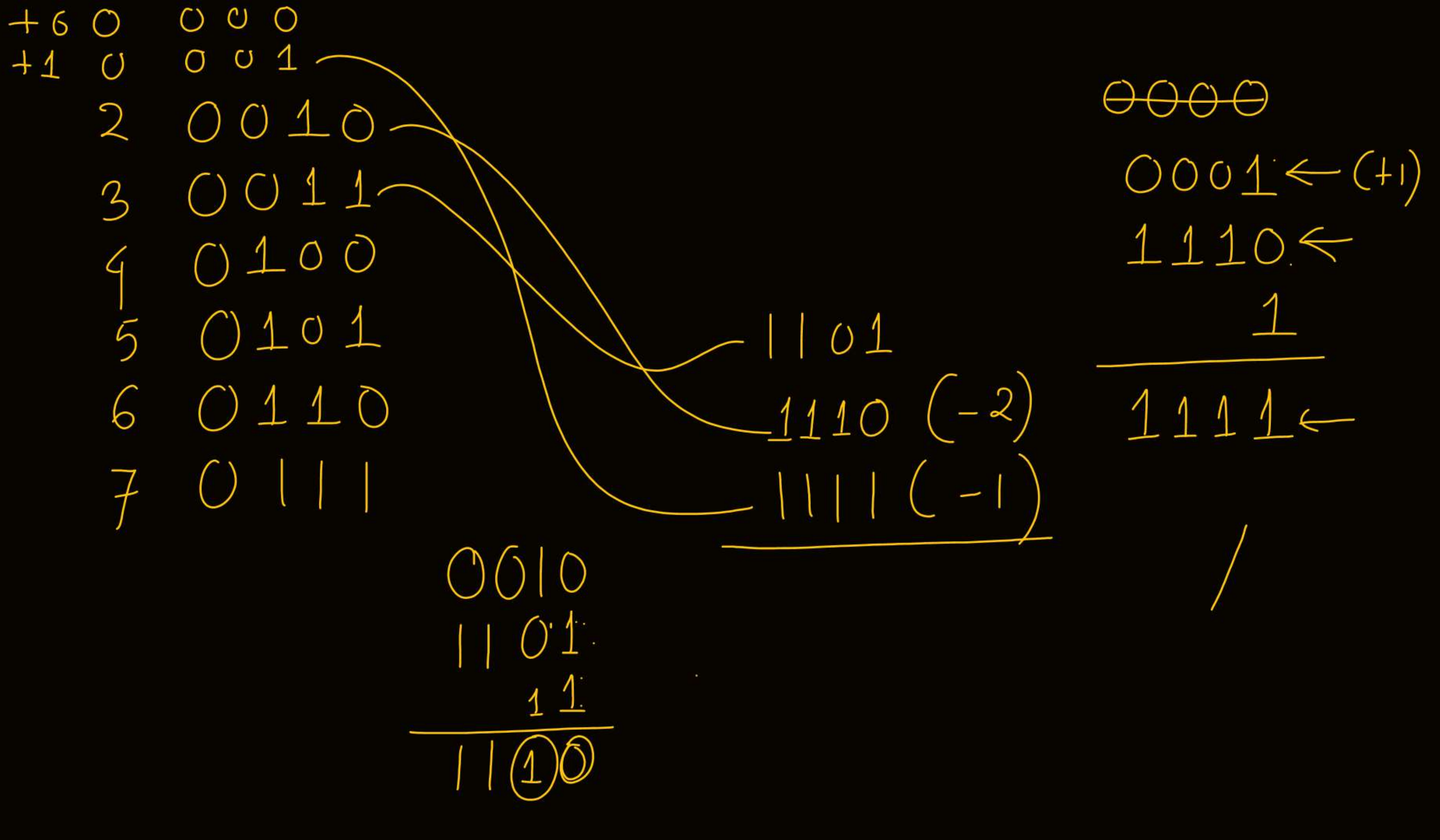
1110

\+ 1

1111 = (-1)


2's complement
0000
0001 (+1)
(+2) 0010
+3 0011
(+4) 0100
0101
0110
0111
+ve
1000
1001 (-7)
1010
1011
1100
1101
1110
1111
-8
-7
(-6)
(-5)
-4
(-3)
(-2)
(-1)
Negative
0001
1110
1
1111
MSB - 1
0010
1101
1
1110
0011 (+3)
1100
+1
1101


t.me/AbhishekKumarpw
Capital
Capital


+6 0 0 0 0
+1 0 0 0 1
2 0010
3 0011
4 0100
5 0101
6 0110
7 0111
1101
1110 (-2)
1111 (-1)
0010
1101
11
1100
0000
0001 ← (+1)
1110 ←
1
1111 ←

4 bit No. 0 — 7 (+ve)

Negative Range −1

-2^{n-1} to $2^{n-1}-1$

-8

Range of Signed

-2^{n-1} to $2^{n-1}-1$

~~Unsie~~

Unsigned — $0 - 2^n - 1$

Signed integer -2^{n-1} to $\neq 2^{n-1} - 1$

$n = 4$ $\quad -2^3$ to $2^3 - 1$

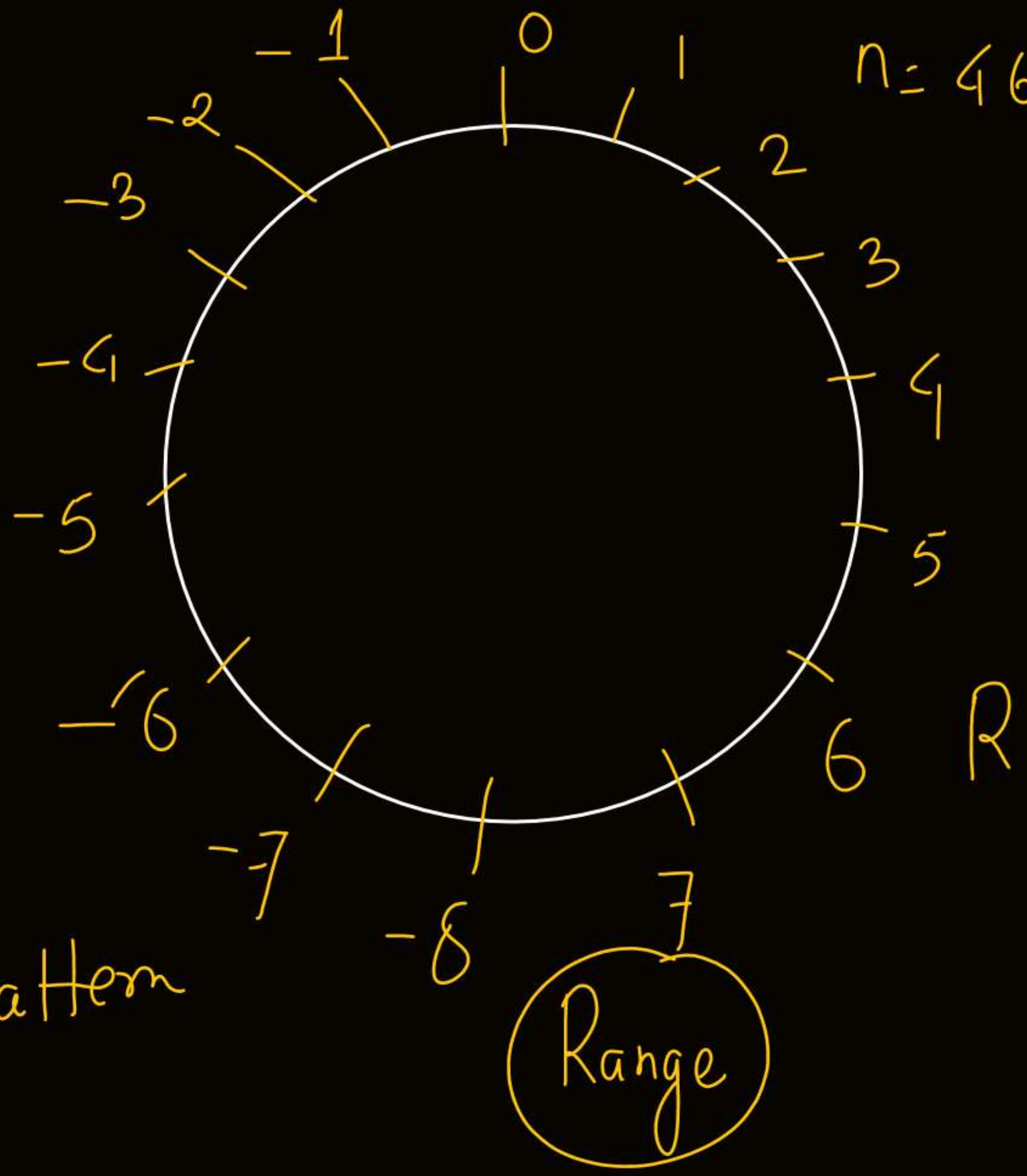
$= -8$ to $+7$

4 bit

a = -15

How to use the range pattern

How to answer overflow pattern



n = 4 bit

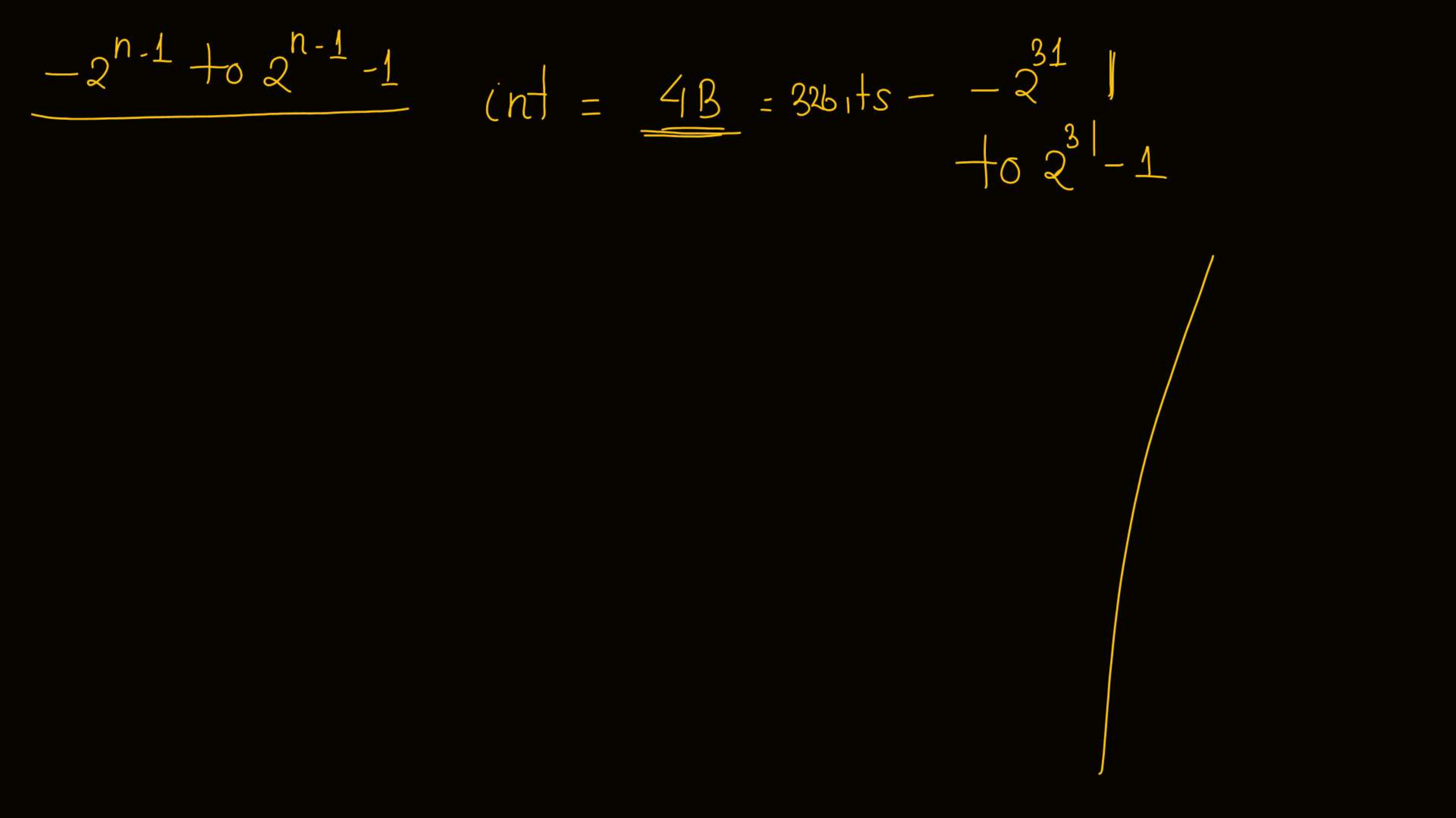
0111

1000

R

-2^{n-1} to $2^{n-1}-1$

int = 4B = 32bits -2^{31} to $2^{31}-1$





Topic : Reserve keyswords

auto	double	int	struct
break	else	long	switch
case	enum	register	typedef
char	extern	return	union
const	float	short	unsigned
continue	for	signed	void
default	goto	sizeof	volatile
do	if	static	while



2 mins Summary

- Topic
- Topic
- Topic
- Topic
- Topic

Slide

THANK - YOU